Series & Binomial Theorem

ANSWER KEY

Section 1: Arithmetic Progressions (AP)

1 An arithmetic progression has first term $a = 7$ and common difference $d = 4$. Find the 20th term u_{20}.

a. **83** ✓

b. 87

c. 79

d. 80

$u_n = a + (n-1)d \Rightarrow u_{20} = 7 + 19(4) = 7 + 76 = 83$.

2 Find the sum of the first 25 terms of the AP: $3, 8, 13, 18, \ldots$

a. **1575** ✓

b. 1625

c. 1500

d. 1550

$a = 3, d = 5$. $S_{25} = \frac{25}{2}[2(3) + 24(5)] = \frac{25}{2}[6 + 120] = \frac{25}{2}(126) = 25 \times 63 = 1575$.

3 In an AP, the 5th term is 18 and the 12th term is 39. Find the first term a and common difference d.

a. $a = 6, d = 3$ ✓

b. $a = 3, d = 3$

c. $a = 9, d = 3$

d. $a = 6, d = 4$

$a + 4d = 18$ and $a + 11d = 39$. Subtracting: $7d = 21 \Rightarrow d = 3$. Then $a = 18 - 4(3) = 6$.

Section 2: Geometric Progressions (GP)

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4 A geometric progression has first term $a = 5$ and common ratio $r = 2$. Find the 8th term u_8.

a. **640** ✓

b. 1280

c. 320

d. 512

$u_n = ar^{n-1} \Rightarrow u_8 = 5(2)^7 = 5(128) = 640.$

5 Find the sum of the first 6 terms of the GP: $3, -6, 12, -24, \ldots$

a. -63 ✓

b. 63

c. -127

d. -65

$a = 3, r = -2. S_6 = \frac{3(1-(-2)^6)}{1-(-2)} = \frac{3(1-64)}{3} = 1 - 64 = -63.$

6 The 2nd term of a GP is 12 and the 5th term is 324. Find the common ratio r.

a. 3 ✓

b. 2

c. 4

d. $\sqrt{3}$

$\frac{u_5}{u_2} = \frac{ar^4}{ar} = r^3 = \frac{324}{12} = 27 \Rightarrow r = 3.$

Section 3: Sum to Infinity of a Convergent GP

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7 State the condition for the sum to infinity of a geometric progression to exist.

a. $|r| < 1$ ✓

b. $r > 1$

c. $r \leq 1$

d. $|r| > 1$

The infinite geometric series converges if and only if the absolute value of the common ratio is strictly less than 1: $|r| < 1$.

8 Find the sum to infinity of the GP: $16, 12, 9, 6.75, \ldots$

a. **64** ✓

b. 48

c. 32

d. 72

$a = 16, r = 12/16 = 3/4$. Since $|3/4| < 1$, $S_\infty = \frac{a}{1-r} = \frac{16}{1-3/4} = \frac{16}{1/4} = 64$.

9 Express the recurring decimal $0.\overline{27} = 0.272727\ldots$ as an exact fraction in lowest terms using an infinite geometric series.

a. $\frac{3}{11}$ ✓

b. $\frac{27}{100}$

c. $\frac{9}{33}$

d. $\frac{7}{25}$

$0.27 + 0.0027 + \ldots$ has $a = 0.27 = 27/100$ and $r = 1/100$. $S_\infty = \frac{27/100}{1-1/100} = \frac{27/100}{99/100} = \frac{27}{99} = \frac{3}{11}$.

Section 4: The Binomial Theorem for Positive Integers

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10 Find the coefficient of x^3 in the expansion of $(2+3x)^5$.

a. 1080 ✓

b. 720

c. 540

d. 1440

General term is $\binom{5}{r}2^{5-r}(3x)^r$. For x^3, $r=3$: $\binom{5}{3}2^2(3)^3 = 10\times 4\times 27 = 1080$.

11 Find the term independent of x in the expansion of $\left(x^2+\dfrac{2}{x}\right)^6$.

a. **240** ✓

b. 60

c. 160

d. 120

General term is $\binom{6}{r}(x^2)^{6-r}\left(\frac{2}{x}\right)^r = \binom{6}{r}2^r x^{12-2r-r} = \binom{6}{r}2^r x^{12-3r}$. Independent when $12-3r=0 \Rightarrow r=4$. Term is $\binom{6}{4}2^4 = 15\times 16 = 240$.

12 In the expansion of $(1+ax)^6$, the coefficient of x^2 is 60. Given $a>0$, find a.

a. **2** ✓

b. 4

c. 3

d. 5

Coefficient of x^2 is $\binom{6}{2}a^2 = 15a^2$. $15a^2 = 60 \Rightarrow a^2 = 4 \Rightarrow a = 2$ (since $a>0$).

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